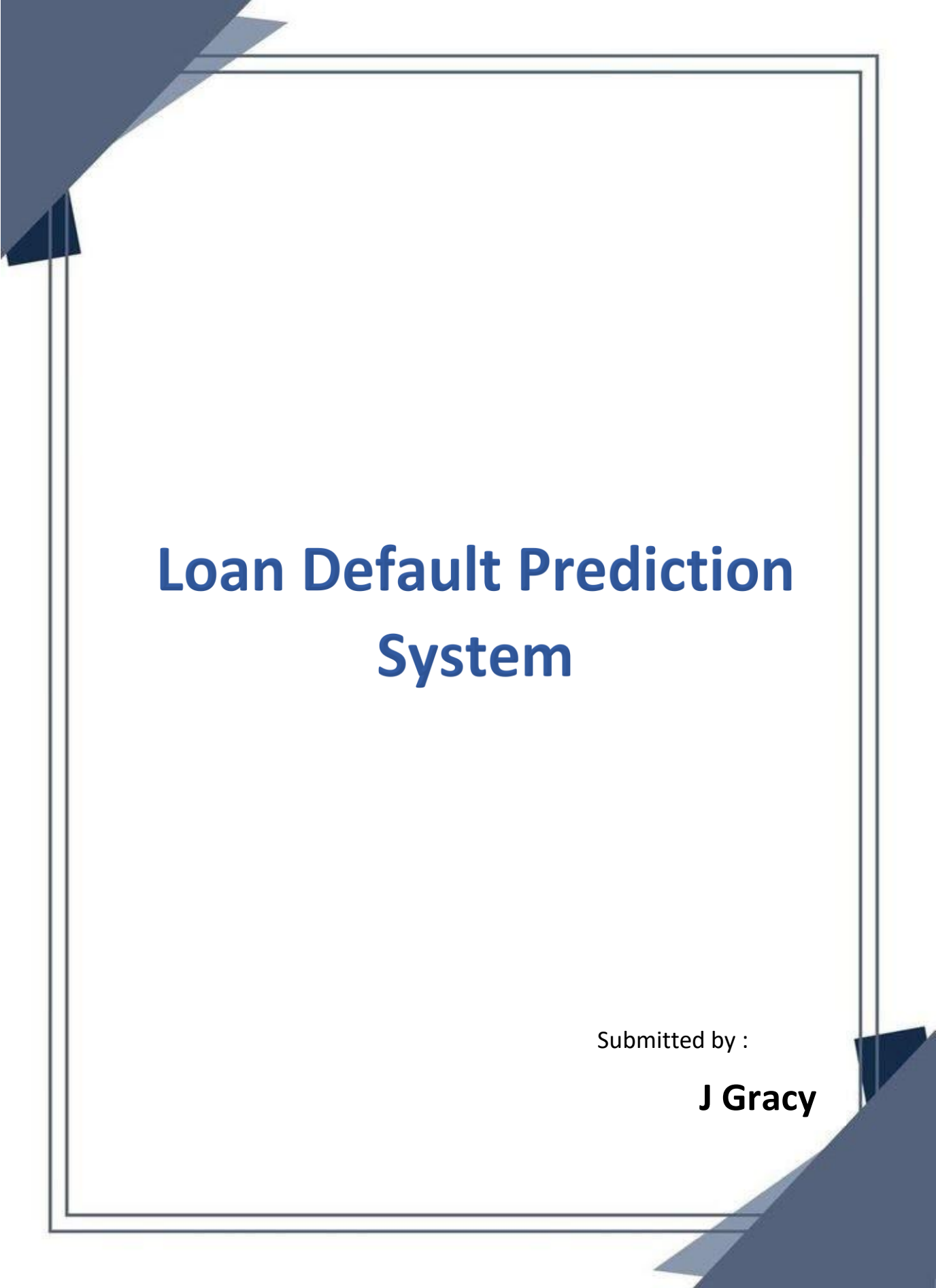




Loan Default Prediction System

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1. Introduction

Financial institutions receive thousands of loan applications every year. One of the biggest challenges faced by banks and lending organizations is determining whether a borrower will repay the loan or default. Incorrect lending decisions can lead to financial losses, making loan default prediction an important problem in the banking and finance sector.

The **Loan Default Prediction System** is a machine learning project developed to predict whether a customer is likely to default on a loan based on financial and demographic information. The project analyzes customer attributes such as age, income, credit score, employment type, education level, loan amount, loan purpose, marital status, and other related factors to identify patterns associated with loan default.

The project follows a complete data science workflow that includes data collection, data preprocessing, exploratory data analysis (EDA), feature engineering, machine learning model development, SQL-based data analysis, and interactive dashboard creation using Power BI. Three machine learning algorithms—**Logistic Regression**, **Decision Tree**, and **Random Forest**—were implemented and compared to evaluate their predictive performance.

In addition to predictive modeling, SQL was used to perform data analysis and retrieve meaningful information from the dataset, while Power BI was used to develop an interactive dashboard for visualizing key performance indicators and business insights. The project demonstrates how machine learning and business intelligence tools can support data-driven decision-making in the financial sector.

2. Objective

The primary objective of this project is to develop a machine learning-based system that can accurately predict whether a customer is likely to default on a loan using historical financial and demographic data. The project aims to assist financial institutions in making informed lending decisions and minimizing the risk associated with loan defaults.

The specific objectives of this project are:

- To collect and analyze the Loan Default Prediction dataset.
- To preprocess and clean the dataset by handling missing values and preparing it for machine learning.
- To perform Exploratory Data Analysis (EDA) to understand the relationships between different customer attributes.
- To build and compare multiple machine learning models, including Logistic Regression, Decision Tree, and Random Forest.
- To evaluate model performance using appropriate evaluation metrics.
- To perform SQL-based data analysis for extracting meaningful insights from the dataset.
- To design an interactive Power BI dashboard for visualizing important business insights and key performance indicators.
- To identify the most influential factors contributing to loan default prediction.

3. Methodology

The Loan Default Prediction System was developed by following a systematic data science workflow. The project involved data collection, preprocessing, exploratory data analysis, machine learning model development, SQL-based analysis, and visualization using Power BI. Each stage contributed to improving the accuracy and interpretability of the prediction system.

The methodology followed in this project consists of the following steps:

3.1 Data Collection

The dataset used for this project contains customer information such as age, income, credit score, employment type, education level, marital status, loan amount, loan purpose, and loan default status. The dataset was obtained from Kaggle and imported into Python for further analysis.

3.2 Data Preprocessing

The dataset was cleaned by checking for missing values, removing unnecessary columns, handling duplicate records, converting categorical variables into numerical form, and preparing the data for machine learning algorithms.

3.3 Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset and identify important patterns. Various charts and statistical summaries were used to analyze relationships between different variables such as income, loan amount, education level, employment type, and loan default status.

3.4 Machine Learning Model Development

The processed dataset was divided into training and testing sets. Three machine learning algorithms—Logistic Regression, Decision Tree, and Random Forest—were trained and evaluated. Their performance was compared using appropriate evaluation metrics to identify the best-performing model.

3.5 SQL Data Analysis

The cleaned dataset was imported into SQLite, where SQL queries were executed to extract meaningful insights. Operations such as filtering, grouping, aggregation, sorting, and subqueries were used to analyze the data.

3.6 Power BI Dashboard

The final stage involved creating an interactive Power BI dashboard to visualize important insights. KPI cards, charts, and slicers were designed to help users understand loan distribution, customer demographics, default rates, and other key metrics.

4. Code and Implementation Details

4.1 Development Tools

Tool	Purpose
Python	Data preprocessing and machine learning
Colab Notebook	Code development and execution
Pandas	Data manipulation
NumPy	Numerical computations
Matplotlib & Seaborn	Data visualization
Scikit-learn	Machine learning algorithms
SQLite	SQL query execution
DB Browser for SQLite	Database management
Power BI Desktop	Dashboard creation

4.2 Dataset Description

The project uses the Loan Default Prediction dataset obtained from Kaggle. The dataset contains customer demographic and financial information used to predict whether a customer will default on a loan. Important attributes include:

- Loan ID
- Age
- Income
- Credit Score
- Employment Type
- Education
- Marital Status
- Loan Amount
- Loan Purpose
- Default (Target Variable)

```
# Load the loan default dataset into a pandas DataFrame
df = pd.read_csv('/content/Loan_default.csv')

# Display the first five records
df.head()
```

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	EmploymentType	MaritalStatus	HasMortgage	HasDependents	LoanPurpose
0	I38PQUQS96	56	85994	50587	520	80	4	15.23	36	0.44	Bachelor's	Full-time	Divorced	Yes	Yes	Cash
1	HPSK72WA7R	69	50432	124440	458	15	1	4.81	60	0.68	Master's	Full-time	Married	No	No	Cash
2	C1OZ6DPJ8Y	46	84208	129188	451	26	3	21.17	24	0.31	Master's	Unemployed	Divorced	Yes	Yes	Cash
3	V2KKSFM3UN	32	31713	44799	743	0	3	7.07	24	0.23	High School	Full-time	Married	No	No	Business
4	EY08JDHTZP	60	20437	9139	633	8	4	6.51	48	0.73	Bachelor's	Unemployed	Divorced	No	Yes	Cash

4.3 Data Preprocessing

Data preprocessing is one of the most important stages of any machine learning project because the quality of the input data directly affects the performance of the model. Before training the algorithms, the dataset was carefully examined and cleaned to ensure that it was accurate, consistent, and suitable for analysis.

Initially, the dataset was imported into Python using the Pandas library. Basic information about the dataset, such as the number of rows, columns, data types, and summary statistics, was examined to understand its structure. The dataset

was then checked for missing values, duplicate records, and inconsistent data entries.

Categorical attributes such as **Education**, **Employment Type**, **Marital Status**, and **Loan Purpose** were converted into numerical values using encoding techniques so that they could be processed by machine learning algorithms. Irrelevant or unnecessary attributes were removed, and the remaining features were selected based on their importance to loan default prediction.

Finally, the dataset was divided into training and testing datasets. The training dataset was used to build the machine learning models, while the testing dataset was used to evaluate their performance on unseen data.

Operations performed during preprocessing:

- Imported the dataset into Python.
- Checked data types and dataset structure.
- Identified and handled missing values.
- Removed duplicate records.
- Encoded categorical variables into numerical form.
- Selected relevant features for prediction.
- Split the dataset into training and testing sets.

```
# Display Dataset Information
df.info()

.. <class 'pandas.core.frame.DataFrame'>
   RangeIndex: 255347 entries, 0 to 255346
   Data columns (total 18 columns):
    #   Column                Non-Null Count  Dtype
   ---  ---
    0   LoanID                 255347 non-null object
    1   Age                   255347 non-null int64
    2   Income                 255347 non-null int64
    3   LoanAmount             255347 non-null int64
    4   CreditScore            255347 non-null int64
    5   MonthsEmployed         255347 non-null int64
    6   NumCreditLines         255347 non-null int64
    7   InterestRate           255347 non-null float64
    8   LoanTerm               255347 non-null int64
    9   DTIRatio              255347 non-null float64
    10  Education              255347 non-null object
    11  EmploymentType         255347 non-null object
    12  MaritalStatus          255347 non-null object
    13  HasMortgage            255347 non-null object
    14  HasDependents          255347 non-null object
    15  LoanPurpose            255347 non-null object
    16  HasCoSigner            255347 non-null object
    17  Default                255347 non-null int64
   dtypes: float64(2), int64(8), object(8)
   memory usage: 35.1+ MB
```

```
# Check For Missing Values
df.isnull().sum()
```

	0
LoanID	0
Age	0
Income	0
LoanAmount	0
CreditScore	0
MonthsEmployed	0
NumCreditLines	0
InterestRate	0
LoanTerm	0
DTIRatio	0
Education	0
EmploymentType	0
MaritalStatus	0
HasMortgage	0
HasDependents	0
LoanPurpose	0
HasCoSigner	0
Default	0

dtype: int64

```
# Check For Duplicate Records
df.duplicated().sum()

np.int64(0)
```

```
# Display the count of each class
df['Default'].value_counts()
```

	count
Default	
0	225694
1	29653

dtype: int64

4.4 Machine Learning Model Development

After preprocessing the dataset, machine learning models were developed to predict whether a customer is likely to default on a loan. The prepared dataset was divided into training and testing datasets using an 80:20 ratio. The training dataset was used to build the models, while the testing dataset was used to evaluate their predictive performance.

Three classification algorithms were implemented and compared:

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. In this project, it was used as a baseline model to predict whether a customer would default on a loan. The algorithm estimates the probability of loan default based on the relationship between the input features and the target variable.

```
from sklearn.linear_model import LogisticRegression

log_reg = LogisticRegression(
    random_state=42,
    max_iter=1000
)

log_reg.fit(X_train, y_train)

...
LogisticRegression(max_iter=1000, random_state=42)

# -----
# 9.Predict Loan Default
# -----

# Predict the class labels
y_pred_lr = log_reg.predict(X_test)

# Predict the probability of loan default
y_prob_lr = log_reg.predict_proba(X_test)[:, 1]

# -----
# Import Evaluation Metrics
# -----

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    roc_auc_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay,
    RocCurveDisplay
)
```

```
# -----
# Logistic Regression Performance
# -----

accuracy = accuracy_score(y_test, y_pred_lr)
precision = precision_score(y_test, y_pred_lr)
recall = recall_score(y_test, y_pred_lr)
f1 = f1_score(y_test, y_pred_lr)
roc_auc = roc_auc_score(y_test, y_prob_lr)

print(f"Accuracy : {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall   : {recall:.4f}")
print(f"F1 Score  : {f1:.4f}")
print(f"ROC-AUC   : {roc_auc:.4f}")
```

```
Accuracy : 0.8817
Precision: 0.7179
Recall    : 0.0358
F1 Score  : 0.0682
ROC-AUC   : 0.7465
```

```
▶ # -----
# Classification Report
# -----

print(classification_report(y_test, y_pred_lr))
```

```
...
              precision    recall  f1-score   support

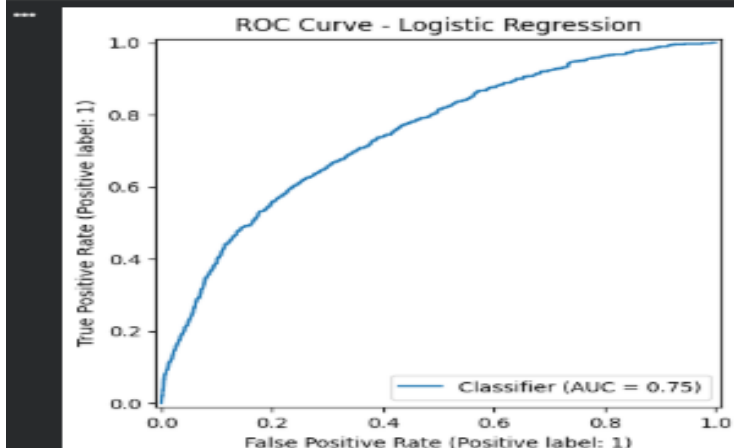
     0.0         0.88        1.00        0.94         5687
     1.0         0.72        0.04        0.07          782

 accuracy         0.88         0.52         0.50         6469
  macro avg       0.80         0.52         0.50         6469
 weighted avg     0.86         0.88         0.83         6469
```

```
▶ # -----
# ROC Curve
# -----

RocCurveDisplay.from_predictions(
    y_test,
    y_prob_lr
)

plt.title("ROC Curve - Logistic Regression")
plt.show()
```



Decision Tree

Decision Tree is a classification algorithm that predicts outcomes by splitting the dataset into branches based on different feature values. It is easy to interpret and helps identify the most influential factors affecting loan default prediction.

```
# -----
# 10.Import Decision Tree Classifier
# -----

from sklearn.tree import DecisionTreeClassifier

# -----
# Train the Decision Tree Model
# -----

dt_model = DecisionTreeClassifier(
    random_state=42
)

dt_model.fit(X_train, y_train)

# -----
# Predict using Decision Tree
# -----

y_pred_dt = dt_model.predict(X_test)
y_prob_dt = dt_model.predict_proba(X_test)[:,1]

# -----
# Decision Tree Performance
# -----

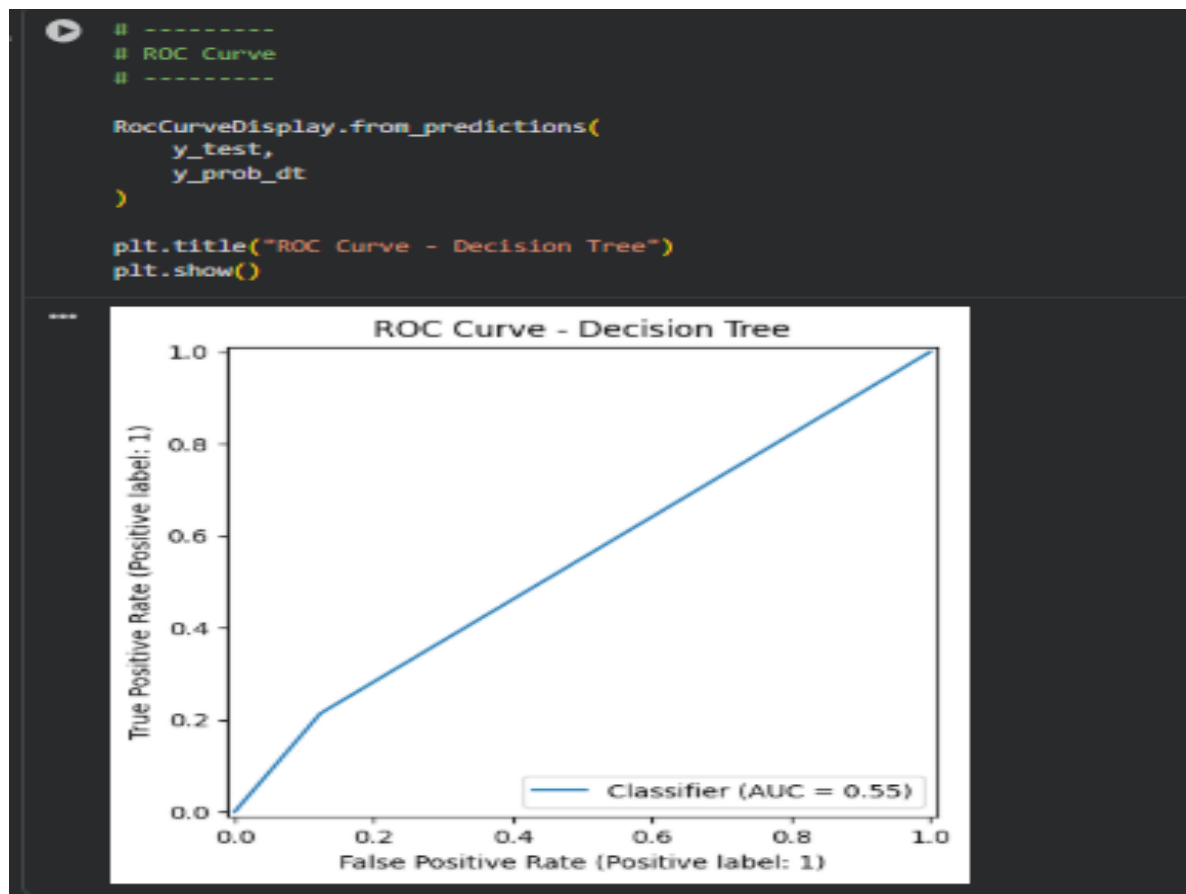
accuracy_dt = accuracy_score(y_test, y_pred_dt)
precision_dt = precision_score(y_test, y_pred_dt)
recall_dt = recall_score(y_test, y_pred_dt)
f1_dt = f1_score(y_test, y_pred_dt)
roc_auc_dt = roc_auc_score(y_test, y_prob_dt)

print(f"Accuracy : {accuracy_dt:.4f}")
print(f"Precision: {precision_dt:.4f}")
print(f"Recall   : {recall_dt:.4f}")
print(f"F1 Score  : {f1_dt:.4f}")
print(f"ROC-AUC   : {roc_auc_dt:.4f}")

*** Accuracy : 0.7967
    Precision: 0.1926
    Recall   : 0.2136
    F1 Score  : 0.2025
    ROC-AUC   : 0.5452
```

```
# -----  
# Classification Report  
# -----  
  
print(classification_report(y_test, y_pred_dt))
```

	precision	recall	f1-score	support
0.0	0.89	0.88	0.88	5687
1.0	0.19	0.21	0.20	782
accuracy			0.80	6469
macro avg	0.54	0.55	0.54	6469
weighted avg	0.81	0.80	0.80	6469



Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. The final prediction is obtained by taking the majority vote from all the individual trees, resulting in a more robust and reliable model.

```
# -----
# 11.Import Random Forest Classifier
# -----

from sklearn.ensemble import RandomForestClassifier
```

```
# -----
# Train the Random Forest Model
# -----
```

```
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)
```

```
> RandomForestClassifier
RandomForestClassifier(random_state=42)
```

```
# -----
# Predict using Random Forest
# -----
```

```
y_pred_rf = rf_model.predict(X_test)
y_prob_rf = rf_model.predict_proba(X_test)[:,1]
```

```
# -----
# Random Forest Performance
# -----
```

```
accuracy_rf = accuracy_score(y_test, y_pred_rf)
precision_rf = precision_score(y_test, y_pred_rf)
recall_rf = recall_score(y_test, y_pred_rf)
f1_rf = f1_score(y_test, y_pred_rf)
roc_auc_rf = roc_auc_score(y_test, y_prob_rf)

print(f"Accuracy : {accuracy_rf:.4f}")
print(f"Precision: {precision_rf:.4f}")
print(f"Recall : {recall_rf:.4f}")
print(f"F1 Score : {f1_rf:.4f}")
print(f"ROC-AUC : {roc_auc_rf:.4f}")
```

```
--- Accuracy : 0.8804
Precision: 0.7222
Recall : 0.0166
F1 Score : 0.0325
ROC-AUC : 0.7173
```

```
# -----
# Classification Report
# -----
```

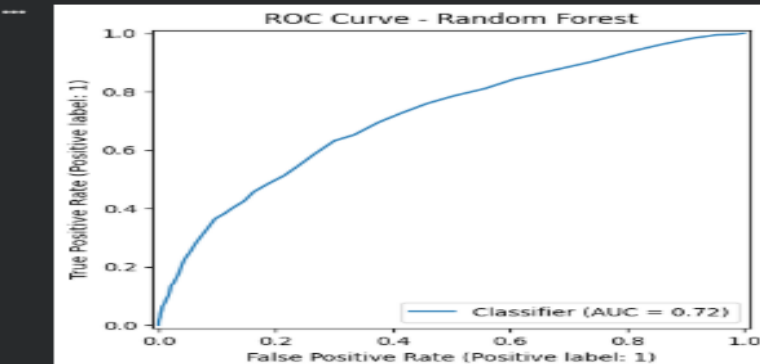
```
print(classification_report(y_test, y_pred_rf))
```

	precision	recall	f1-score	support
0.0	0.88	1.00	0.94	5687
1.0	0.72	0.02	0.03	782
accuracy			0.88	6469
macro avg	0.80	0.51	0.48	6469
weighted avg	0.86	0.88	0.83	6469

```
# -----
# ROC Curve
# -----
```

```
RocCurveDisplay.from_predictions(
    y_test,
    y_prob_rf
)
```

```
plt.title("ROC Curve - Random Forest")
plt.show()
```



Model Evaluation

The performance of all three models was evaluated using appropriate classification metrics such as **Accuracy**, **Precision**, **Recall**, **F1-Score**, and the **Confusion Matrix**. Comparing these metrics helped determine which model performed best for loan default prediction.

```
# -----  
# 12.Model Comparison  
# -----  
  
comparison = pd.DataFrame({  
    "Model": [  
        "Logistic Regression",  
        "Decision Tree",  
        "Random Forest"  
    ],  
    "Accuracy": [  
        accuracy,  
        accuracy_dt,  
        accuracy_rf  
    ],  
    "Precision": [  
        precision,  
        precision_dt,  
        precision_rf  
    ],  
    "Recall": [  
        recall,  
        recall_dt,  
        recall_rf  
    ],  
    "F1 Score": [  
        f1,  
        f1_dt,  
        f1_rf  
    ],  
    "ROC-AUC": [  
        roc_auc,  
        roc_auc_dt,  
        roc_auc_rf  
    ]  
})  
  
comparison
```

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.881744	0.717949	0.035808	0.068210	0.748495
1	Decision Tree	0.796723	0.192618	0.213555	0.202547	0.545234
2	Random Forest	0.880352	0.722222	0.016624	0.032500	0.717294

Next steps: [Generate code with comparison](#) [New interactive sheet](#)

```
comparison.style.highlight_max(  
    subset=["Accuracy", "Precision", "Recall", "F1 Score", "ROC-AUC"],  
    color="pink"  
)
```

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.881744	0.717949	0.035808	0.068210	0.748495
1	Decision Tree	0.796723	0.192618	0.213555	0.202547	0.545234
2	Random Forest	0.880352	0.722222	0.016624	0.032500	0.717294

4.5 Sample Prediction

After training the machine learning model, a sample customer record was provided as input to verify the prediction capability of the system. The model analyzed the customer's financial and demographic information and predicted whether the customer was likely to default on the loan.

This step demonstrates the practical application of the developed model, showing how it can be used to evaluate new loan applicants and support decision-making in financial institutions.

```
# =====  
# Predict Loan Default for a New Applicant  
# =====  
  
new_customer = pd.DataFrame({  
    'Age': [35],  
    'Income': [70000],  
    'LoanAmount': [25000],  
    'CreditScore': [720],  
    'MonthsEmployed': [60],  
    'NumCreditlines': [3],  
    'InterestRate': [8.5],  
    'LoanTerm': [36],  
    'DTIRatio': [0.25],  
    'Education': ['Bachelor'],  
    'EmploymentType': ['Full-time'],  
    'MaritalStatus': ['Married'],  
    'HasMortgage': ['Yes'],  
    'HasDependents': ['No'],  
    'LoanPurpose': ['Home'],  
    'HasCoSigner': ['Yes']  
})  
  
# Convert categorical variables to dummy variables  
new_customer = pd.get_dummies(new_customer)  
  
# Match the training dataset columns  
new_customer = new_customer.reindex(columns=X.columns, fill_value=0)  
  
# Scale the applicant's features  
new_customer_scaled = scaler.transform(new_customer)  
  
# Predict loan default  
prediction = rf_model.predict(new_customer_scaled)  
  
# Predict probability  
probability = rf_model.predict_proba(new_customer_scaled)  
  
print("-" * 45)  
print("      LOAN DEFAULT PREDICTION")  
print("-" * 45)  
  
if prediction[0] == 1:  
    print("Prediction          : Default")  
else:  
    print("Prediction          : No Default")  
  
print(f"Probability of Default: {probability[0][1]*100:.2f}%")  
print(f"Probability of Repayment: {probability[0][0]*100:.2f}%")  
  
print("-" * 45)
```

```
*** -----  
      LOAN DEFAULT PREDICTION  
-----  
Prediction          : No Default  
Probability of Default: 1.00%  
Probability of Repayment: 99.00%  
-----
```

4.6 SQL Implementation

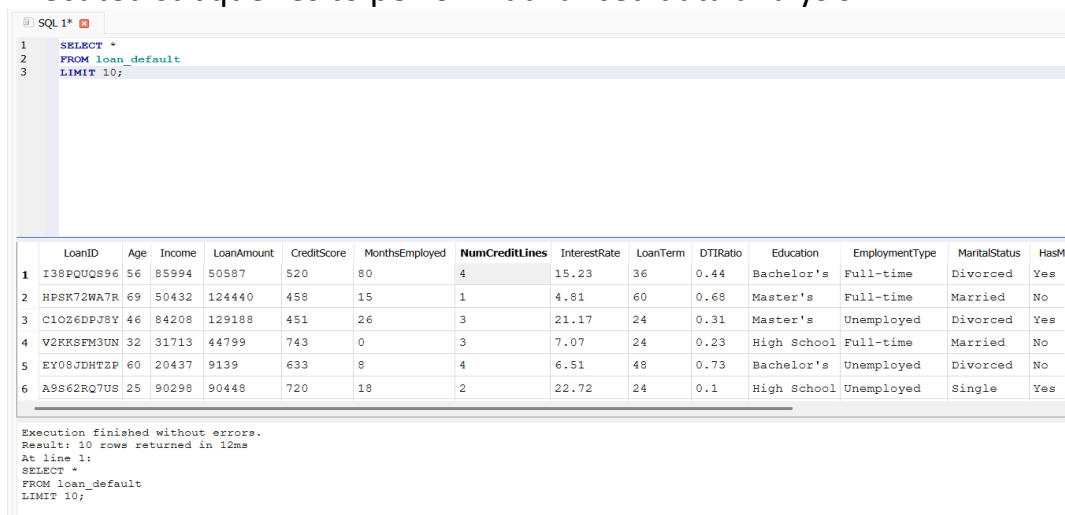
To perform database analysis, the cleaned loan default dataset was imported into **SQLite** using **DB Browser for SQLite**. SQL (Structured Query Language) was used to retrieve, filter, summarize, and analyze the data efficiently. This helped in understanding customer characteristics and identifying important patterns related to loan defaults.

Several SQL queries were executed using different SQL clauses, including **SELECT**, **WHERE**, **GROUP BY**, **ORDER BY**, **COUNT**, **AVG**, **MIN**, **MAX**, and **subqueries**. These queries provided valuable insights into customer income, loan amounts, employment types, education levels, and default distribution.

The SQL implementation demonstrated how relational databases can be used alongside machine learning to organize, query, and analyze large datasets efficiently.

SQL Operations Performed

- Imported the dataset into SQLite.
- Retrieved records using **SELECT** statements.
- Filtered data using the **WHERE** clause.
- Grouped records using **GROUP BY**.
- Sorted results using **ORDER BY**.
- Calculated statistical values using aggregate functions such as **COUNT**, **AVG**, **MIN**, and **MAX**.
- Executed subqueries to perform advanced data analysis.



The screenshot shows the DB Browser for SQLite interface. The top pane contains a SQL query: `SELECT * FROM loan_default LIMIT 10;`. The bottom pane displays the results of this query as a table with 14 columns and 10 rows of data.

	LoanID	Age	Income	LoanAmount	CreditScore	MonthsEmployed	NumCreditLines	InterestRate	LoanTerm	DTIRatio	Education	EmploymentType	MaritalStatus	HasM
1	I38PQUQS96	56	85994	50587	520	80	4	15.23	36	0.44	Bachelor's	Full-time	Divorced	Yes
2	HPSK72WA7R	69	50432	124440	458	15	1	4.81	60	0.68	Master's	Full-time	Married	No
3	C1O26DPJ8Y	46	84208	129188	451	26	3	21.17	24	0.31	Master's	Unemployed	Divorced	Yes
4	V2KKSFM3UN	32	31713	44799	743	0	3	7.07	24	0.23	High School	Full-time	Married	No
5	EY08JDHTZP	60	20437	9139	633	8	4	6.51	48	0.73	Bachelor's	Unemployed	Divorced	No
6	A9S62RQ7US	25	90298	90448	720	18	2	22.72	24	0.1	High School	Unemployed	Single	Yes

Execution finished without errors.
Result: 10 rows returned in 12ms
At line 1:
SELECT *
FROM loan_default
LIMIT 10;

SQL 1*

```
1 SELECT AVG(LoanAmount) AS Average_Loan_Amount
2 FROM loan_default;
```

	Average_Loan_Amount
1	127578.86551242

SQL 1*

```
1 SELECT Education,
2       COUNT(*) AS Total_Customers
3 FROM loan_default
4 GROUP BY Education
5 ORDER BY Total_Customers DESC;
```

	Education	Total_Customers
1	Bachelor's	64366
2	High School	63903
3	Master's	63541
4	PhD	63537

SQL 1*

```
1 SELECT EmploymentType,  
2        SUM("Default") AS Total_Defaults  
3 FROM loan_default  
4 GROUP BY EmploymentType  
5 ORDER BY Total_Defaults DESC;
```

	EmploymentType	Total_Defaults
1	Unemployed	8650
2	Part-time	7677
3	Self-employed	7302
4	Full-time	6024

SQL 1*

```
1 SELECT LoanPurpose,  
2        AVG(InterestRate) AS Average_Interest_Rate  
3 FROM loan_default  
4 GROUP BY LoanPurpose  
5 ORDER BY Average_Interest_Rate DESC;
```

	LoanPurpose	Average_Interest_Rate
1	Home	13.5269305463479
2	Education	13.5126054308401
3	Business	13.4791481149363
4	Other	13.4777691794006
5	Auto	13.4671967193769

4.7 Power BI Dashboard

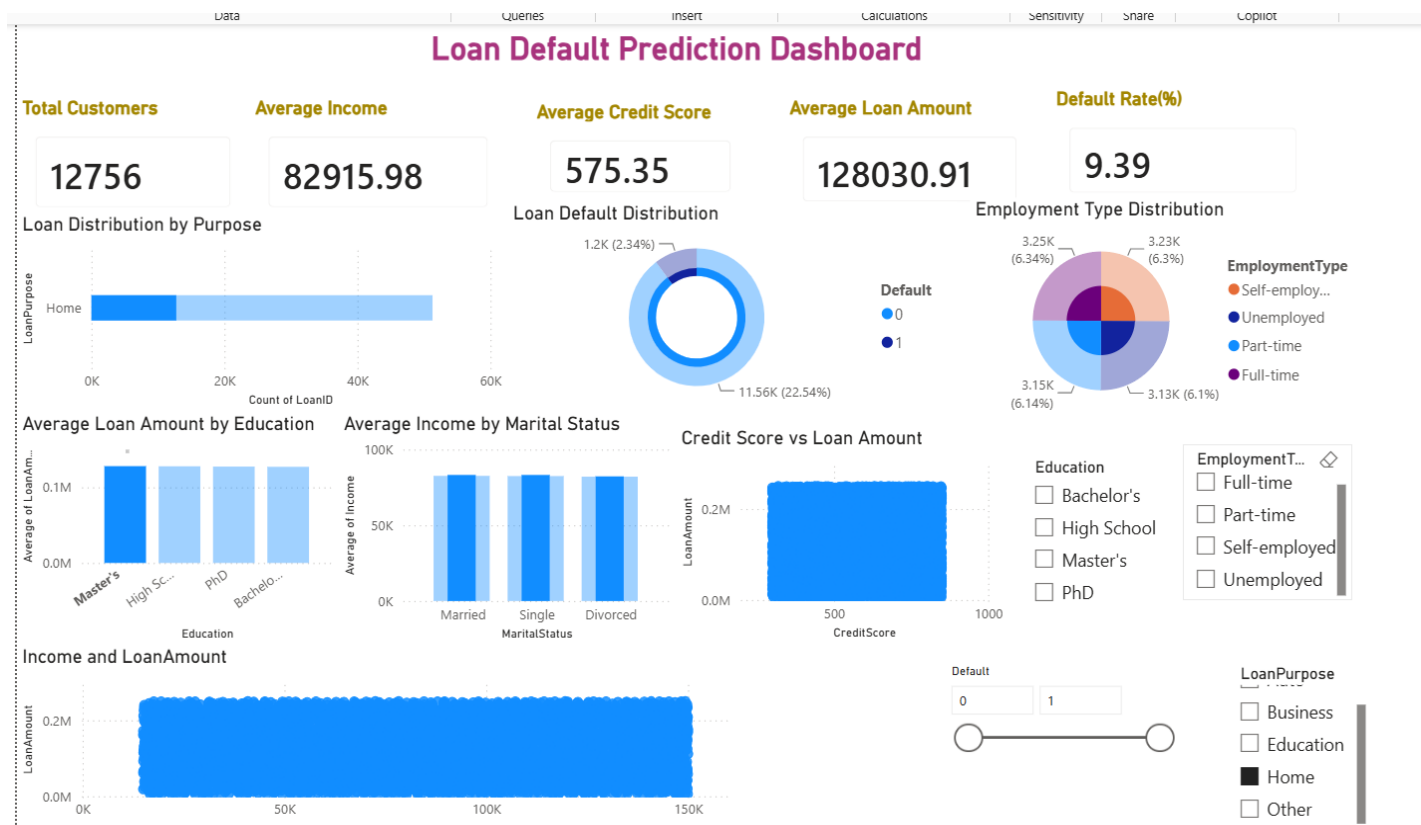
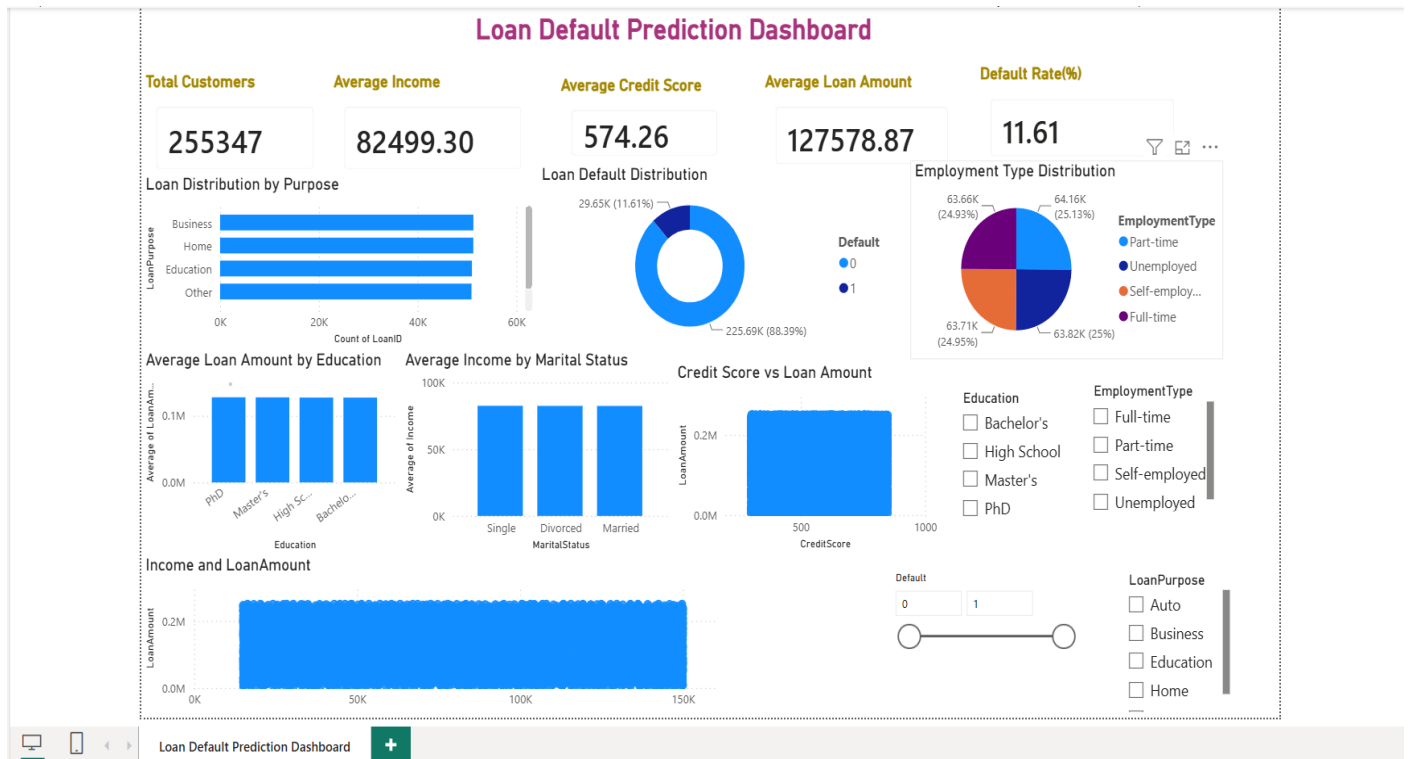
After completing the machine learning and SQL analysis, an interactive dashboard was developed using Microsoft Power BI. The dashboard provides a visual representation of the loan default dataset, allowing users to analyze important information through charts, graphs, KPI cards, and filters.

The dashboard was designed to present key business insights in a simple and interactive manner. Various visualizations were created to display customer demographics, loan distribution, credit score analysis, income distribution, and default trends. Slicers were also added to allow users to filter the data based on different customer attributes.

The dashboard includes the following visual components:

- KPI Cards displaying Total Customers, Average Income, Average Credit Score, Average Loan Amount, and Default Rate.
- Bar charts showing the distribution of loans by purpose.
- Pie and donut charts representing loan default status and employment type.
- Scatter plots illustrating the relationship between customer income, credit score, and loan amount.
- Slicers for filtering data by Education, Employment Type, Loan Purpose, and Default Status.

The interactive dashboard enables users to quickly explore patterns in the dataset and supports better understanding of the factors associated with loan default prediction.



5. Results and Observations

The **Loan Default Prediction System** was successfully developed and implemented using Python, SQLite, and Microsoft Power BI. The project followed a complete data science workflow, beginning with data collection and preprocessing, followed by exploratory data analysis, machine learning model development, SQL-based data analysis, and dashboard visualization.

The dataset was successfully cleaned and prepared for model training by handling missing values, removing inconsistencies, and encoding categorical variables. Three machine learning algorithms—**Logistic Regression, Decision Tree, and Random Forest**—were implemented and evaluated to predict whether a customer is likely to default on a loan. The trained models demonstrated satisfactory performance in classifying customers into default and non-default categories based on their demographic and financial information.

SQL was used to analyze the processed dataset by executing various queries involving filtering, grouping, sorting, and aggregate functions. These queries provided meaningful insights into customer income, loan amounts, employment types, education levels, and loan default trends.

An interactive dashboard was developed using Microsoft Power BI to present the analytical results in a visually appealing and user-friendly format. The dashboard included KPI cards, bar charts, pie charts, scatter plots, and slicers, enabling users to explore the dataset interactively and understand important business insights.

Key Observations

- Customers with higher credit scores generally exhibited a lower probability of loan default.
- Customer income showed a positive relationship with the approved loan amount.
- Employment type and education level influenced customer loan profiles and repayment patterns.
- SQL queries efficiently summarized large volumes of data and extracted valuable business insights.

- The Power BI dashboard enabled interactive analysis through filters, KPI cards, and visualizations.
- The developed machine learning models successfully predicted loan default for new customer records, demonstrating the practical applicability of the system.

5.1 Business Questions Answered

The developed **Loan Default Prediction System** provides valuable business insights that can support financial institutions in making informed lending decisions. Using machine learning models, SQL analysis, and an interactive Power BI dashboard, the following business questions were addressed.

1. How many customers were analyzed?

The dataset consisted of **255,347 customer records**, providing a large and diverse dataset for analysis and model training.

2. What is the average income of customers?

The average annual income of customers was **82,499.30**, which helps financial institutions understand the overall financial profile of their customer base.

3. What is the average credit score of customers?

The average credit score of customers was **574.26**, indicating the overall creditworthiness of the borrowers included in the dataset.

4. What is the average loan amount sanctioned?

The average loan amount was **127,578.87**, providing an overview of the typical loan value issued to customers.

5. What is the overall loan default rate?

The analysis showed an overall loan default rate of **11.61%**, meaning approximately one out of every nine customers defaulted on their loan.

6. How are loan applications distributed across different loan purposes?

The dashboard presents the distribution of loan applications across various purposes such as **Business, Home, Education, and Other**, enabling financial institutions to identify the major borrowing categories and understand customer demand.

7. What is the distribution of borrowers based on employment type?

Borrowers were fairly evenly distributed among **Part-time, Full-time, Self-employed, and Unemployed categories**, allowing comparison of borrowing patterns across different employment groups.

8. Can the system predict whether a new customer will default?

Yes. The developed machine learning models successfully predict whether a customer is likely to default on a loan using demographic and financial attributes. This helps financial institutions assess credit risk before approving loans.

6. Conclusion

The **Loan Default Prediction System** was successfully designed and implemented using **Python, SQLite, and Microsoft Power BI**. The project followed a complete data science workflow, including data preprocessing, exploratory data analysis, machine learning model development, SQL-based data analysis, and interactive dashboard visualization.

Three machine learning algorithms—**Logistic Regression, Decision Tree, and Random Forest**—were implemented to predict whether a customer is likely to default on a loan. The trained models successfully classified customers based on their demographic and financial information, demonstrating the practical application of machine learning in credit risk assessment.

In addition to predictive modeling, SQL was used to perform efficient data querying and analysis, while the Power BI dashboard transformed the analytical results into meaningful visualizations. The dashboard provided key performance indicators and answered important business questions related to customer income, credit scores, loan amounts, employment types, and loan default rates,

enabling better understanding of customer behavior and supporting informed business decisions.

Overall, this project enhanced practical knowledge of data preprocessing, machine learning, SQL, and business intelligence tools. It provided valuable hands-on experience in developing an end-to-end data science solution and demonstrated how predictive analytics can support financial institutions in reducing loan default risk and improving lending decisions.

7.Dataset Link and Files Used

<https://www.kaggle.com/datasets/nikhil1e9/loan-default>

- Kaggle. *Loan Default Prediction Dataset*.
<https://www.kaggle.com/datasets/nikhil1e9/loan-default>
- Python Software Foundation.
<https://www.python.org>
- Scikit-learn Documentation.
<https://scikit-learn.org>
- SQLite Documentation.
<https://www.sqlite.org>
- Microsoft Power BI Documentation.
<https://learn.microsoft.com/power-bi>